

Chapter 6 was amended by [Local Law 77 of 2023](#). This law has an effective date of June 10, 2023.

CHAPTER 6

TYPES OF CONSTRUCTION

SECTION BC 601 GENERAL

601.1 Scope. The provisions of this chapter shall control the classification of buildings as to type of construction.

**TABLE 601
FIRE-RESISTANCE RATING REQUIREMENTS FOR BUILDING ELEMENTS (HOURS)**

BUILDING ELEMENT	TYPE I		TYPE II		TYPE III		TYPE IV	TYPE V ⁱ	
	A	B	A	B	A	B	HT	A	B
Primary structural frame ^f (see Section 202)	3 ^a	2 ^{a,j}	1 ^j	0	1	0	HT	1	0
Bearing walls									
Exterior ^{e, f, g}	3	2	1	0	2	2	2	1	0
Interior	3 ^a	2 ^a	1	0	1	0	1/HT	1	0
Nonbearing walls and partitions	See Table 602								
Exterior									
Nonbearing walls and partitions							See Section 602.4.6		
Interior ^d	0	0	0	0	0	0		0	0
Floor construction ^h and associated secondary members (see Section 202)	2	2	1	0	1	0	HT	1	0
Roof construction and associated secondary members (see Section 202)	1 1/2 ^{b, c}	1 ^{b, c}	1 ^{b, c}	0 ^c	1 ^{b, c}	0	HT	1 ^{b, c}	0

For SI: 1 foot = 304.8 mm.

- a. Roof supports: Fire-resistance ratings of primary structural frame and bearing walls are permitted to be reduced by 1 hour where supporting a roof only.
- b. 1. Except in Group F-1, H, M and S-1 occupancies, fire protection of structural members in roof construction shall not be required, including protection of primary structural frame members within any portion of the roof, roof framing and decking where every part of the roof construction is 20 feet or more above any floor immediately below. Fire-retardant-treated wood members shall be allowed to be used for such unprotected members.
2. Except in Group F occupancies subject to regulation under Sections 264(1) and 264(2) of the *New York State Labor Law*, and in Group I-1, R-1, and R-2 occupancies, in Types I and II construction, fire-retardant-treated wood shall be allowed in buildings including girders and trusses as part of the roof construction when the building is:
 - i. Type II construction of any height; or
 - ii. Type I construction two stories or less; or when over two stories, the vertical distance from the upper floor to the roof is 20 feet or more.
- c. Except in Group F occupancies subject to regulation under Sections 264(1) and 264(2) of the *New York State Labor Law*, and in Group I-1, R-1 and R-2 occupancies, heavy timber shall be allowed where a 1-hour or less fire-resistance rating is required.
- d. Not less than the fire-resistance rating required by other sections of this code.
- e. Not less than the fire-resistance rating based on fire separation distance (see Table 602).
- f. Not less than the fire-resistance rating as referenced in Section 704.10.
- g. See note g of Table 602.
- h. See Section 711.2.4 for additional requirements.
- i. Type V construction is not permitted inside fire districts except as provided for in Section D105.1 of Appendix D.
- j. See Section 403.2.1 for additional requirements for high-rise buildings and fire-resistance rating of columns in such buildings.

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SECTION BC 602 CONSTRUCTION CLASSIFICATION

602.1 General. Buildings and structures erected or to be erected, altered or extended in height or area shall be classified in one of the five construction types defined in Sections 602.2 through 602.5. The building elements shall have a fire-resistance rating not less than that specified in Table 601 and exterior walls shall have a fire-resistance rating not less than that specified in Table 602. Where required to have a fire-resistance rating by Table 601, building elements shall comply with the applicable provisions of Section 703.2. The protection of openings, ducts and air transfer openings in building elements shall not be required unless required by other provisions of this code. Buildings constructed or altered inside the fire district shall further comply with Appendix D.

TABLE 602
FIRE-RESISTANCE RATING REQUIREMENTS FOR EXTERIOR WALLS BASED ON FIRE SEPARATION DISTANCE^{a, e, g, h, j}

FIRE SEPARATION DISTANCE = X (feet)	TYPE OF CONSTRUCTION	OCCUPANCY GROUP H ^f	OCCUPANCY GROUP F-1, M, S-1 ^{g, i}	OCCUPANCY GROUP A, B, E, F-2, I, R, S-2, U ^b
$X < 5^c$	All	3	2	1
$5 \leq X < 10$	IA	3	2	1
	Others	2	1	1
$10 \leq X < 30$	IA, IB	2	1	1 ^d
	IIB, VB	1	0	0
	Others	1	1	1 ^d
$X \geq 30$	All	0	0	0

For SI: 1 foot = 304.8 mm.

- Load-bearing exterior walls shall also comply with the fire-resistance rating requirements of Table 601.
- Group U when used as accessory to Group R-3 shall not be required to have a fire-resistance rating where the fire separation distance is 5 feet or more for free standing private garages in compliance with Section 406.1, and when the separation distance is 3 feet or more for other freestanding Group U buildings. For free standing private garages where the fire separation distance is less than 5 feet, refer to Section 406.3.4 for required fire-resistance rating for exterior walls.
- See Section 706.1.1 for party walls.
- Open parking garages complying with Section 406 shall not be required to have a fire-resistance rating.
- The fire-resistance rating of an exterior wall is determined based upon the fire separation distance of the exterior wall and the story in which the wall is located.
- For special requirements for Group H occupancies, see Section 415.6.
- Inside the fire district, exterior load-bearing walls of Type II buildings shall have a fire-resistance rating not less than prescribed below:

$X < 5$	2 hours
$5 \leq X < 10$	2 hours
$10 \leq X < 30$	1 hour
$X \geq 30$	As per Table 602.
- Inside the fire district, exterior nonload-bearing walls of Type II buildings shall have a fire-resistance rating not less than prescribed below:

$X < 5$	As per Table 602
$5 \leq X < 10$	As per Table 602
$10 \leq X < 30$	1 hour
$X \geq 30$	As per Table 602.
- For special requirements for Group S aircraft hangars, see Section 412.4.1.
- Where Table 705.8 permits unbearing exterior walls with unlimited area of unprotected openings, the required fire-resistance rating for the exterior walls is 0 hours.

602.1.1 Minimum requirements. A building or portion thereof shall not be required to conform to the details of a type of construction higher than that type which meets the minimum requirements based on occupancy even though certain features of such a building actually conform to a higher type of construction. Classification shall be that of the minimum requirement unless all of the requirements for the higher type of construction are met.

Exception: Portions of buildings that cantilever over an adjacent building or tax lot shall also comply with the fire-resistance ratings of Section 705.12.

602.2 Types I and II. Types I and II construction are those types of construction in which the building elements listed in Table 601 are of noncombustible materials, except as permitted in Section 603 and elsewhere in this code.

602.3 Type III. Type III construction is that type of construction in which the exterior walls are of noncombustible materials and the interior building elements are of any material permitted by this code. Fire-retardant-treated wood framing complying with Section 2303.2 shall be permitted within exterior wall assemblies of a 2-hour rating or less.

Exceptions:

1. In Group I-1, R-1, and R-2 occupancies, all exterior walls, fire walls, exit passageways, and shaft enclosures shall be noncombustible.
2. In Group F occupancies subject to Section 270(1) of the *New York State Labor Law*, all exterior wall assemblies and all structural elements shall meet the requirements for a “fireproof building” as such term is defined in Section 264 of such law.
3. Inside the fire district, exterior load-bearing walls shall be constructed of noncombustible material.
4. Inside the fire district, exterior nonload-bearing walls may be constructed with fire-retardant-treated wood complying with Section 2303.2 where the building is equipped throughout with an automatic sprinkler system in accordance with Sections 903.3.1.1 through 903.3.1.3, unless otherwise prohibited by Exception 1 or 2 above.

‡‡‡ **602.4 Type IV.** Type IV construction is that type of construction in which the exterior walls are of noncombustible materials or other materials permitted by Section 602.4.1 or 602.4.2, and the interior building elements are of solid wood, glue-laminated timber, heavy timber (HT), structural composite lumber (SCL), or cross-laminated timber (CLT) without concealed spaces. The minimum dimensions for permitted materials including solid timber, glued-laminated timber, SCL, and CLT and details of Type IV construction shall comply with the provisions of Section 2304.11 and this section. Interior walls and partitions not less than 1-hour fire-resistance rating or heavy timber complying with Section 2304.11.2.2 shall be permitted. Buildings of Type IV construction utilizing SCL or CLT shall be equipped throughout with an automatic sprinkler system where required by Section 903.2.13. In buildings of Type IV construction utilizing SCL or CLT, a fire watch shall be maintained in accordance with Section 901.7.2 of the *New York City Fire Code* and Section 3303.3 of this code.

Exceptions:

1. In Group I-1, R-1, and R-2 occupancies, all exterior walls, fire walls, exit passageways, and shaft enclosures shall be noncombustible.
2. In Group F occupancies subject to Section 270(1) of the *New York State Labor Law*, all exterior wall assemblies and all structural elements shall meet the requirements for a “fireproof building” as defined in Section 264 of such law.
3. Inside the fire district, exterior load-bearing walls shall be constructed of noncombustible material.
4. Inside the fire district, exterior nonbearing walls may be constructed with fire-retardant-treated wood complying with Section 2303.2 of this code where the building is equipped throughout with an automatic sprinkler system in accordance with Sections 903.3.1.1 through 903.3.1.3, unless otherwise prohibited by Exception 1 or 2 above.
5. Inside the fire district, exterior nonbearing walls are permitted to be constructed with cross-laminated timber (CLT) complying with Section 602.4.2 of this code, unless otherwise prohibited by Exception 1 or 2 above.

602.4.1 Fire-retardant-treated wood in exterior walls. Fire-retardant-treated wood framing and sheathing complying with Section 2303.2 shall be permitted within exterior wall assemblies not less than 6 inches (152.4 mm) in thickness with a 2-hour rating or less.

602.4.2 Cross-laminated timber in exterior walls. Cross-laminated timber (CLT) complying with Section 2303.1.4 shall be permitted within exterior wall assemblies not less than 6 inches (152.4 mm) in thickness with a 2-hour rating or less, provided the exterior surface of the cross-laminated timber (CLT) is protected by one of the following:

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1. Fire-retardant-treated wood sheathing complying with Section 2303.2 and not less than $1\frac{5}{32}$ inch (11.9 mm) thick;
2. Type X gypsum board not less than $\frac{5}{8}$ inch (15.9 mm) thick; or
3. A noncombustible material.

602.4.3 Exterior structural members. Where a horizontal separation of 20 feet (6096 mm) or more is provided, wood columns and arches conforming to heavy timber sizes in accordance with Section 2304.11 shall be permitted to be used externally.

Exception: Where prohibited by Exceptions 1, 2 or 3 of Section 602.4.

602.5 Type V. Type V construction is that type of construction in which the structural elements, exterior walls and interior walls are of any materials permitted by this code. Type V construction shall not be permitted inside the fire district unless otherwise permitted by Section D105.1.

Exception: In Group F occupancies subject to Section 270(1) of the *New York State Labor Law*, all exterior wall assemblies and all structural elements shall meet the requirements for a “fireproof building” as defined in Section 264 of such law.

SECTION BC 603 COMBUSTIBLE MATERIAL IN TYPES I AND II CONSTRUCTION

603.1 Allowable materials. Combustible materials shall be permitted in buildings of Type I or II construction in the following applications and in accordance with Sections 603.1.1 through 603.1.3:

1. Fire-retardant-treated wood, complying with Section 2303.2, shall be permitted in:
 - 1.1. Nonbearing interior partitions where the required fire-resistance rating is 1 hour or less.

Exception: Public corridors and exits shall be constructed of noncombustible materials.
 - 1.2. Roof construction as permitted in Table 601, Note b.
2. Thermal and acoustical insulation, other than foam plastics, having a flame spread index of not more than 25.

Exceptions:

1. Insulation placed between two layers of noncombustible materials without an intervening airspace shall be allowed to have a flame spread index of not more than 100.
2. Insulation installed between a finished floor and solid decking without intervening airspace shall be allowed to have a flame spread index of not more than 200.
3. Foam plastics in accordance with Chapter 26.
4. Roof coverings that have an A or B classification as defined in Section 1505.
5. Interior floor finish and floor covering materials installed in accordance with Section 804.
6. Millwork such as doors, door frames, window sashes and frames.
7. Interior wall and ceiling finishes installed in accordance with Sections 801 and 803.
8. Trim installed in accordance with Section 806.
9. Where not installed greater than 15 feet (4572 mm) above grade, show windows, nailing or furring strips and wooden bulkheads below show windows, including their frames, aprons and show cases.
10. Finish flooring installed in accordance with Section 805.
11. Partitions dividing portions of stores, offices or similar places occupied by one tenant only and that do not establish a corridor serving an occupant load of 30 or more shall be permitted to be constructed of fire-retardant-treated wood, 1-hour fire-resistance-rated construction or of wood panels or similar light construction up to 6 feet (1828.8 mm) in height.
12. Stages and platforms constructed in accordance with Sections 410.3 and 410.4, respectively.

13. Combustible exterior wall coverings, balconies and similar projections and bay or oriel windows in accordance with Chapter 14.
14. Blocking such as for handrails, millwork, cabinets and window and door frames.
15. Light-transmitting plastics as permitted by Chapter 26.
16. Mastics and caulking materials applied to provide flexible seals between components of exterior wall construction.
17. Exterior plastic veneer installed in accordance with Section 2605.2.
18. Nailing or furring strips as permitted by Section 803.13.
19. Heavy timber as permitted by Note c to Table 601 and Section 602.4.3.
20. Aggregates, component materials and admixtures as permitted by Section 703.2.2.
21. Sprayed fire-resistant materials and intumescent and mastic fire-resistant coatings, determined on the basis of fire resistance tests in accordance with Section 703.2 and installed in accordance with Sections 1705.14 and 1705.15, respectively.
22. Materials used to protect penetrations in fire-resistance-rated assemblies in accordance with Section 714.
23. Materials used to protect joints in fire-resistance-rated assemblies in accordance with Section 715.
24. Materials allowed in the concealed spaces of buildings of Types I and II construction in accordance with Section 718.5.
25. Materials exposed within plenums complying with Section 602 of the *New York City Mechanical Code*.
26. Bleachers, grandstands, and folding and telescopic seating within an assembly space in accordance with Section 1029.1.1 of this code.

603.1.1 Ducts. The use of nonmetallic ducts shall be permitted where installed in accordance with the limitations of the *New York City Mechanical Code*.

603.1.2 Piping. The use of combustible piping materials shall be permitted where installed in accordance with the limitations of the *New York City Mechanical Code* and the *New York City Plumbing Code*.

603.1.3 Electrical. The use of electrical wiring methods with combustible insulation, tubing, raceways and related components shall be permitted where installed in accordance with the limitations of the *New York City Electrical Code*.